

Topic 13 — Electromagnetic Induction

Edexcel GCSE (9-1) Physics (1PH0) — Paper 2

13.1 Inducing an EMF

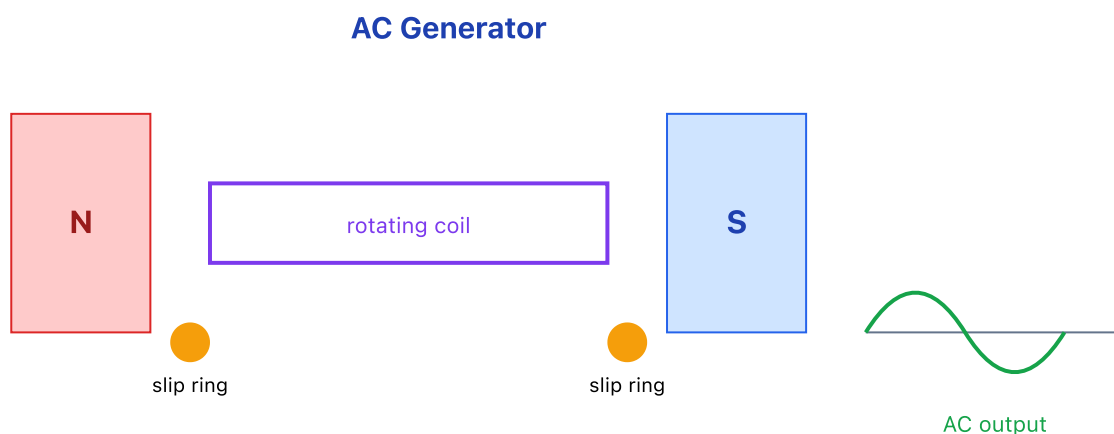
Moving a conductor through a magnetic field, or changing the magnetic field through a coil, induces an **EMF (potential difference)**. If part of a circuit, an induced current will flow.

Ways to increase induced EMF:

- Move the magnet (or wire) faster;
- Use a stronger magnet;
- Increase the number of turns on the coil;
- Use a soft iron core.

13.2 Generators (Alternators)

A coil rotates in a magnetic field; the changing flux induces an alternating EMF. A simple AC generator uses **slip rings** and brushes; the output is AC.



13.3 Transformers

A transformer changes the voltage of an **alternating** supply. It has a primary coil and a secondary coil, both wound on a soft iron core. Alternating current in the primary creates a changing magnetic flux, inducing an AC in the secondary.

$$V_s / V_p = N_s / N_p$$

$$V_p I_p = V_s I_s \quad (\text{ideal transformer — power in = power out})$$

Step-up vs Step-down

- **Step-up:** more turns on secondary → higher V_s , lower I_s .
- **Step-down:** fewer turns on secondary → lower V_s , higher I_s .

Worked Example: Primary 230 V, $N_p = 100$, $N_s = 1000$.

$$V_s = 230 \times 1000/100 = \mathbf{2300 \text{ V}}.$$

$$\text{If } I_p = 4 \text{ A, } I_s = 230 \times 4 / 2300 = \mathbf{0.40 \text{ A}}.$$

13.4 National Grid and Transmission Losses

Transmission lines dissipate power as I^2R . Doubling the voltage halves the current for the same power → quarters the heat loss. Hence very high voltages (up to 400 kV) are used, then stepped down for safe use.

Exam Tip: Transformers need changing current — they don't work with steady DC because a constant flux induces no EMF.

13.5 Everyday Uses

- Microphones: sound vibrates a diaphragm/coil in a field → induced EMF.
- Dynamos and alternators on bicycles / in cars.
- Induction cooking, wireless phone charging, electric toothbrushes.

Key Points Summary

- Induced EMF requires a *change* in flux through a coil.
- Increase EMF by faster movement, stronger magnet, more turns, iron core.
- Generators produce AC using slip rings; DC dynamos use a commutator.
- Transformers: $V_s/V_p = N_s/N_p$; power conserved (ideal).
- National Grid: step up for transmission (low $I \rightarrow$ low I^2R loss), step down near towns.